

PIX IRT Results Summary

Automated Experiment Report

March 5, 2026

Setup

All runs use student-holdout split (80/20), full dataset, and online test-student state updates. Main dataset: `pix_mapping/pix_irt_outcome.csv`.

Embedding Dimension Grid (MIRT)

Best test AUC per embedding dimension from a 5-epoch grid:

emb_dim	best_epoch	train_AUC	test_AUC	train_Acc	test_Acc
2	1	0.7805	0.7760	0.7029	0.6993
3	1	0.7843	0.7757	0.7065	0.6985
5	1	0.7915	0.7750	0.7130	0.6989
8	1	0.8028	0.7739	0.7230	0.6965

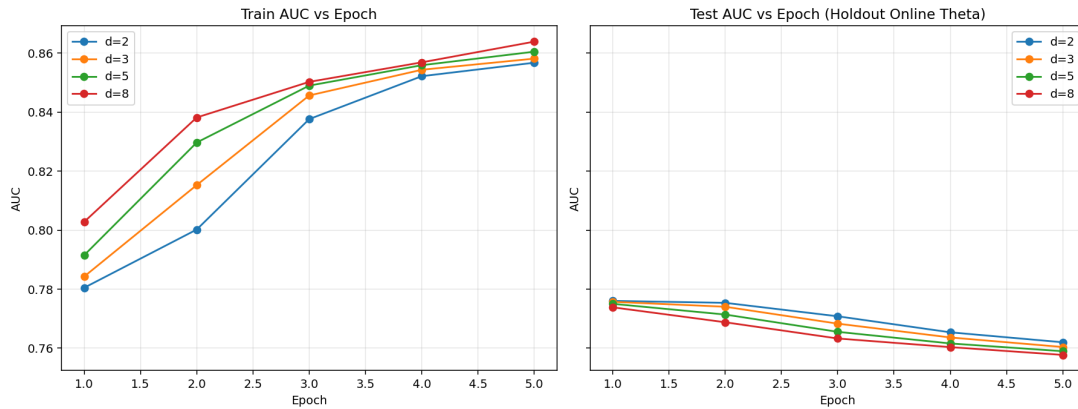


Figure 1: Train/Test AUC curves for `emb_dim` in $\{2,3,5,8\}$.

Dim=2 Hyperparameter Sweep

Sweep over training hyperparameters (`lr`, `student_l2`), ranked by best test AUC:

lr	student_l2	best_epoch	best_test_AUC	best_test_Acc	final_test_AUC	final_train_AUC
0.005	0.001	2	0.7761	0.6992	0.7717	0.8361
0.005	0.010	2	0.7761	0.6991	0.7744	0.8459
0.010	0.001	1	0.7761	0.6991	0.7618	0.8568
0.010	0.010	1	0.7759	0.6989	0.7670	0.8572
0.020	0.001	1	0.7754	0.6985	0.7558	0.8632
0.020	0.010	1	0.7750	0.6982	0.7647	0.8616

Theta Update Step Sweep (Dim=2)

Effect of online update step `theta_lr` on test metrics:

checkpoint_epoch	theta_lr	train_AUC	test_AUC	test_Acc	test_NLL
5	0.30	0.8371	0.7788	0.7001	0.5534
5	0.20	0.8371	0.7777	0.6993	0.5553
2	0.30	0.7820	0.7772	0.7001	0.5505
2	0.20	0.7820	0.7769	0.7000	0.5507
2	0.10	0.7820	0.7762	0.6989	0.5513
2	0.05	0.7820	0.7748	0.6979	0.5526
2	0.02	0.7820	0.7725	0.6955	0.5550
5	0.10	0.8371	0.7720	0.6945	0.5633
5	0.05	0.8371	0.7623	0.6878	0.5783
5	0.02	0.8371	0.7488	0.6780	0.6034

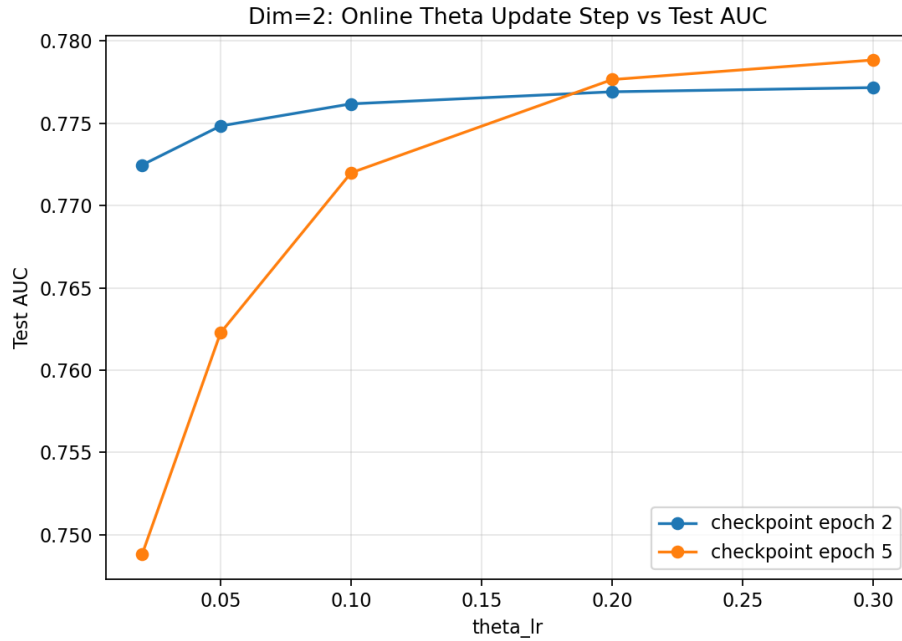


Figure 2: Test AUC vs `theta_lr` at checkpoints epoch 2 and 5 (dim=2).

1D Comparison

Comparison of strict 1PL (Rasch) versus 1D MIRT on the same protocol:

Model	Train Acc	Train AUC	Test Acc	Test AUC
Strict 1PL ($\sigma(\theta - diff)$)	0.7514	0.8340	0.6977	0.7760
1D MIRT (Adam)	0.7357	0.8184	0.7022	0.7798

Item Bias Comparison: Strict 1PL vs MIRT Dim=2

For strict 1PL, comparable bias is defined as $bias_{1PL} = -diff$. For MIRT dim=2, bias is item_bias.

- Pearson correlation: 0.7760
- Spearman correlation: 0.7411
- Mean absolute difference: 0.6257

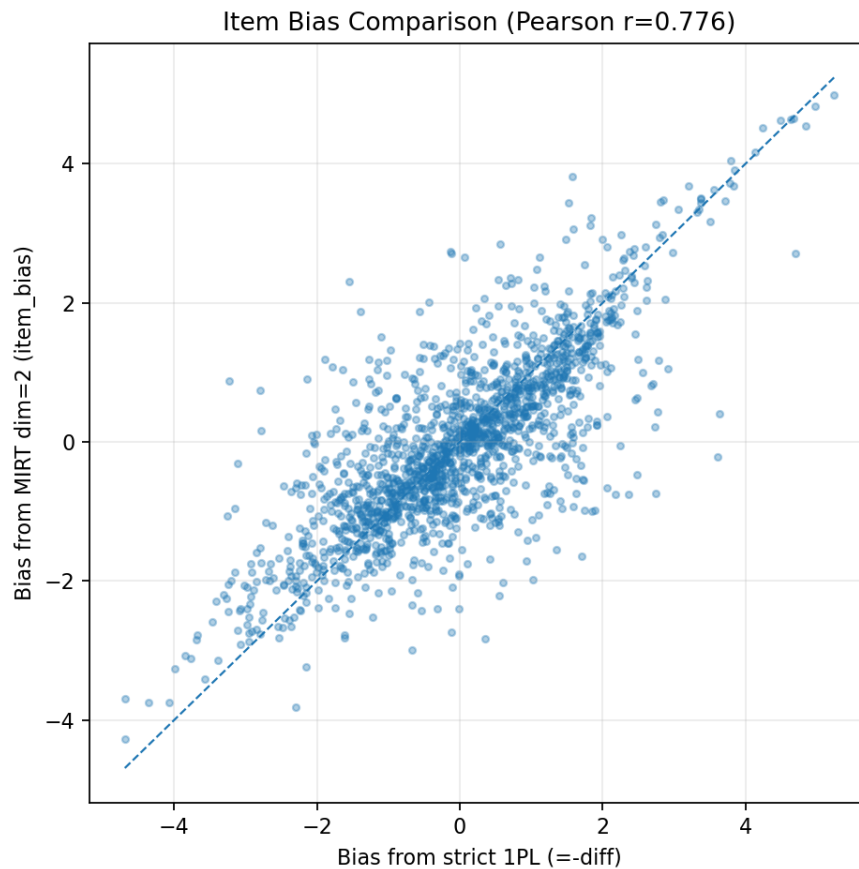


Figure 3: Per-item bias comparison between strict 1PL and MIRT dim=2.

Largest Bias Gaps (Top 10)

item_idx	item_raw	bias_1PL	bias_MIRT2	delta(MIRT2-1PL)
1026	1037	-3.2249	0.8760	4.1009
1472	1478	-1.5517	2.3077	3.8595
1133	1135	3.6049	-0.2183	-3.8232
1410	1416	-2.7955	0.7353	3.5308
1566	1572	2.7419	-0.7395	-3.4814
1246	1249	1.7071	-1.6491	-3.3562
1513	1518	-1.3918	1.8758	3.2677
1243	1245	3.6312	0.4034	-3.2278
1310	1312	0.3592	-2.8315	-3.1907
753	753	2.3643	-0.7598	-3.1241